

Year:I**Semester:I**

Course Code	CourseTitle	Credit Hrs	Lecture (Hrs)	Tutorial(Hrs)	Practical (Hrs)	Total (Hrs)
BT101CO	Introduction to Programming	3	3	1	2	6
BT102CO	Digital Logic	3	3	1	2	6
BT103HS	Calculus	3	3	1		4
BT104HS	Introduction to Artificial Intelligence	3	3	1		4
BT105HS	Technical Report Writing and Presentation	3	3	1		4
BT106HS	Society and Professional Ethics	3	3	1		4
	Total	18	18	6	4	28

Year:I**Semester:II**

Course Code	Course Title	Credit Hrs	Lecture(Hrs)	Tutorial(Hrs)	Laboratory (Hrs)	Total(Hrs)
BT151CO	Object Oriented Programming	3	3	1	2	6
BT152CO	Artificial Intelligence and Intelligent Systems	3	3	1	2	6
BT153CO	Microprocessor and Assembly Language	3	3	1	2	6
BT154CO	Probability and Statistics	3	3	1		4
BT155HS	Linear Algebra	3	3	1		4
BT156CO	Project I	2	1		2	3
	Total	17	16	5	8	29

**Introduction to Programming
BT101CO**

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1	2	Theory	Practical	Theory	Practical	100
			20	20	60	-	

Course Objectives:

The main objective of this course is to familiarize students with the fundamental concepts of Python scripting language so as to enable them to develop small applications using Python programming language.

Course Contents:

Unit 1: Introduction to Python

[4Hrs]

- 1.1 What is Python?
- 1.2 History of Python
- 1.3 Downloading and Installing Python
- 1.4 Running Python
- 1.5 Python Documentation
- 1.6 Comparing Python

Unit 2: Data Types, Input/ Output

[5 Hrs]

- 2.1 Python Character Set
- 2.2 Python Core Data Types
- 2.3 Tokens
- 2.4 Variables
- 2.5 Assignment of values to variables
- 2.6 The input() function
- 2.7 print() function
- 2.8 The eval() function

Unit 3: Operators

[3 Hrs]

- 3.1 Arithmetic Operators
- 3.2 Operator Precedence and Associativity
- 3.3 Bitwise Operator
- 3.4 The Compound Assignment Operator, Boolean Operator, Relational Operator

Unit 4: Control Statements

[7 Hrs]

- 4.1 Decision making statement-if-else,if-elseif-else
- 4.2 Loop Control Statements- while loop, for loop
- 4.3 Continue, assert, pass, return statements

Unit 5: Functions and Lists**[9 Hrs]**

- 5.1 Syntax and basics of a function
- 5.2 Use of a function
- 5.3 Parameters and Arguments in a function
- 5.4 The local and global scope of a variable
- 5.5 Recursive functions
- 5.6 The lambda function
- 5.7 Creating Lists
- 5.8 Accessing the elements of a List
- 5.9 List Slicing
- 5.10 Python in-built functions for Lists
- 5.11 List

Unit 6: Tuples, Sets and Dictionaries**[9 Hrs]**

- 6.1 Creating Tuples
- 6.2 tuple() function, Inbuilt functions for tuples
- 6.3 Indexing and Slicing
- 6.4 Sort tuples, Traverse Tuples from a list
- 6.5 The Zip() function, The Inverse Zip(*) function
- 6.6 Sets, Python Set Class, Set Operations
- 6.7 Dictionaries- Creating a dictionary, Comparing two Dictionaries, Methods of dictionary Class, Traversing dictionaries, Nested dictionaries, Traversing Nested Dictionaries

Unit 7: Files**[8 Hrs]**

- 7.1 Open() to open a file, Writing Text, Closing files
- 7.2 Writing numbers to a file
- 7.3 Reading numbers from a file
- 7.4 appending data
- 7.5 seek() function

Laboratory work:

There shall be laboratory exercises covering all units mentioned in the syllabus.

Reference Books:

1. W. J. Chun, "Core Python Programming", 3rd Edition, Prentice Hall
2. Allen Downey, "Think Python", Green Tea Press.
3. Kenneth A. Lambert, "Fundamentals of Python", Cengage.
4. Ashok Namdev Kamthane, Amit Ashok Kamthane, "Programming and Problem Solving with Python", McGraw Hill Education

**Digital Logic
BT102CO**

Year: I

Semester: I

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1	2	Theory	Practical	Theory	Practical	100
			20	20	60	-	

Course Objective:

The main objective of this course is to familiarize students with the concepts of design and analysis of digital systems and introduce the principles of digital computer organization and design.

Course contents:

Unit 1: Numbersystems

[5hrs]

- 1.1 Introduction
- 1.2 Comparison between analog and digital system
- 1.3 Number system and conversion (Binary, Octal, and Hexadecimal)
- 1.4 Signed and unsigned numbers, fraction conversion and binary arithmetic
- 1.5 Representation of Binary coded decimal, gray code, alphanumeric code and error detection and correction codes

Unit 2: Boolean algebra andlogicgates

[6Hrs]

- 2.1 Introduction to Boolean algebra
- 2.2 Basic theory and properties of Booleanalgebra
- 2.3 Booleanfunctions
- 2.4 Logic gates and operations

Unit 3: Simplification ofBooleanfunctions

[6 Hrs]

- 3.1 K-Map (Two and three variable maps)
- 3.2Product of sums, sum of products
- 3.3 Simplification of NAND and NOR implementation

Unit 4: Combinationallogic

[8 Hrs]

- 4.1Design procedure of Adders and Subtractors
- 4.2Codeconversion, Analysis procedure
- 4.3Multilevel NAND gates
- 4.4Multilevel NOR gates

Unit 5: Combinational logic

[8Hrs]

- 5.1Binary parallel adder

- 5.2 Decimal adder
- 5.3 Magnitude comparator
- 5.4 Decoders
- 5.5 Multiplexers
- 5.6 Read only memory Programmable logic array (PLA)

Unit 6: Sequential logic

[6Hrs]

- 6.1 Difference between sequential and combinational circuit
- 6.2 Introduction and Design procedure of RS, JK, T, D and master-slave flip flops
- 6.3 Design with state equation and state reduction table

Unit 7: Registers and Counters

[6 Hrs]

- 7.1 Introduction
- 7.2 Left and right shift register
- 7.3 Serial in serial out, Serial in parallel out, Parallel in serial out, Parallel in parallel out.
- 7.4 Asynchronous and Synchronous counter
- 7.5 Asynchronous up and down counter
- 7.6 Decade counter, Ring counter
- 7.7 Application of counter

Laboratory Work:

1. Familiarization with logic gates
2. De Morgan's law
3. Multiplexer and de-multiplexer
4. Encoder and decoder
5. Half adder and half subtractor
6. Full adder and full subtractor
7. RS, JK, T, D and master slave flip flops
8. Shift registers, Sequential logic
9. Ripple counters and synchronous counters
10. Simulation using suitable software

Reference Books:

1. Floyd T. L & Jain R. P, "*Digital Fundamentals*", 8th edition
2. Morris Mano, "*Logic & Computer Design Fundamentals*", Pearson education
3. William I, Fletcher, "*An Engineering Approach to Digital Design*", Prentice Hall of India, New Delhi, 1990
4. A.P. Malvino & Jerald A. Brown, "*Digital Computer Electronics*", 1995
5. D. D. Hodegs & H.G. Jackson, "*Analysis & Design of Digital Integrated Circuits*", McGraw Hill, New York, 1983

Calculus

BT103HS

Year: I

Semester: I

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1		Theory	Practical	Theory	Practical	100
			40		60	-	

Course Objective:

The aim of this course is to provide sound knowledge of differential and integral calculus.

Course Contents:

Unit 1: Derivatives and its Applications

[20 Hrs]

- 1.1 General concept of limit, continuity and differentiability of a function with some examples.
- 1.2 Equation of tangent and normal to the curve.
- 1.3 Successive derivatives and application of Leibnitz's theorem.
- 1.4 Extrema of functions of two or three variables.
- 1.5 Evaluate Maclaurin's and Taylor's expansion for variety of functions and use them to solve further advanced problems.
- 1.6 Indeterminant form and limits (Application of L.Hospital's rule).
- 1.7 Curvature (Radius of curvature and chord of curvature).
- 1.8 Partial differentiation of functions, Euler's theorem.

Unit 2: Integral and its Applications

[20 Hrs]

- 2.1 Indefinite integral and Definite integral (Properties of definite integral).
- 2.2 Improper integral.
- 2.2 Reduction formulae and Beta and Gamma functions.
- 2.3 Area, length, volume, and surface area (Cartesian form only).
- 2.4 Multiple integral (Double and Triple integral).
- 2.5 Change of order of integration in double integral.

Unit 3: Solving first order and first degree differential equations

[5 Hrs]

(variable separable method, linear differential equation, Bernoulli's and exact differential equations and its applications)

Reference Books:

- 1 M.B.Singh and B.C. Bajracharya, DIFFERENTIAL CALCULUS, (sukundapustak Bhawan Kathmandu).

- 2 G.B.Thomas and R.L.Finney, calculus Analytic Geometry, Addison Wesley Publishing Company.
- 3 Prof. G.D. Pant and G.S. Shrestha , Integral calculus and differential equations.
- 4 B.S.Grewal, Higher Engineering Mathematics , Khanna publishers.

Introduction to Artificial Intelligence
BT104CO

Year: I

Semester: I

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1	2	Theory	Practical	Theory	Practical	100
			20	20	60	-	

Course Objective:

The main objective of this course is to familiarize students with the basic and conceptual understanding of AI systems and a practical grounding in its concepts.

Course Contents:

Unit 1: Introduction

[6 Hrs]

- 1.1 Definitions of Intelligence and Artificial Intelligence, AI vs Natural Intelligence
- 1.2 Brief history of AI, Turing Test
- 1.3 Foundations of AI and related fields
- 1.4 Goals and Challenges of AI
- 1.5 Applications of AI

Unit 2: Agents and Problem Solving

[11 Hrs]

- 2.1 Introduction to agent and agent environment
- 2.2 Agent architecture
- 2.3 Types and Examples of Agent
- 2.4 PEAS
- 2.5 Defining Problem as a State Space Search
- 2.6 Problem types
- 2.7 Problem formulation, Problem solving agent, Learning agent
- 2.8 Constraint Satisfaction Problem, Cryptarithmic problems
- 2.9 Game Playing

Unit 3: Search Strategies

[10 Hrs]

- 3.1 Searching and its importance
- 3.2 Uninformed Search: Breadth-first search, Depth-first search, Depth-limited search, iterative-deepening search, Uniform-cost search, Bidirectional search, Comparative study of uninformed search techniques
- 3.3 Informed Search: Best-first search, Greedy search, A* search, Hill climbing, Comparative study of informed search techniques

3.4 Adversarial Search: Games and Perfect Games, Min Max problem, Alpha-Beta Pruning

Unit 4: Knowledge Representation

[9 Hrs]

4.1 Knowledge and its types

4.2 Production-rule systems

4.3 Logic, Formal-logic connectives

4.4 Propositional Logic, Truth tables, , Well-formed formula, Tautology

4.5 Predicate Logic, FOPL

4.6 Semantic Nets and Frames

Unit 5: Machine Learning

[9 Hrs]

5.1 Winston's learning concept

5.2 Rote learning, Learning by Analogy, Explanation-based learning, Inductive learning

5.3 Supervised and Unsupervised learning

5.4 Decision Trees

5.5 Reinforcement learning

5.6 Evolutionary learning, Genetic algorithm

5.7 Case Study

Laboratory Work:

There shall be lab exercises on PROLOG/LISP covering the following topics.

1. Simple question-answering
2. Solving family relation problems
3. Numerical problems such as Factorial, Factors, GCD
4. Logic-gate programs
5. Fundamental search techniques

Reference Books:

1. Stuart Russel & Peter Norvig, "*Artificial Intelligence: A Modern Approach*", Pearson
2. E. Rich & K. Knight, "*Artificial Intelligence*", McGraw-Hill
3. P. H. Winston, "*Artificial Intelligence*", Addison-Wesley
4. R. Shingal, "*Formal Concepts in Artificial Intelligence*", Chapman & Hall
5. G. F. Luger & W. A. Stubblefield, "*Artificial Intelligence*", Benjamin Cummings
6. D. Crookes, "*Introduction to Programming in Prolog*", Prentice Hall

Technical Report Writing and Presentation
BT105HS

Year: I

Semester: I

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1		Theory	Practical	Theory	Practical	100
			40		60	-	

Course Objective:

The basic objective of the course is to develop linguistic and communicative competence of students with special emphasis on skill development and English language skill for effective presentation.

Course Contents:

Unit:1 Oral Presentation

[10 hrs]

1.1 Introduction

1.2 Fundamentals of effective oral presentation

Facial expression, Gesture, Eye Contact, Space Distancing, Posture

1.3 Type of communication

Verbal communication, Nonverbal communication

1.4 Dimension of communication

Downward of communication, Upward communication, Horizontal communication

1.5 Activities

Group discussion, Debate, Public speaking

Unit: 2 Reading

[10 Hrs]

2.1 Introduction

2.2 Type of reading

Intensive reading, Extensive reading, Skimming reading, Scanning reading

2.3 Activities

Note taking

Summary/precise writing

2.4 Text for reading

‘A Tale’ by B.P. Koirala

‘Marriage is a Private Affair’ by Chinua Achebe

‘Who was to Blame?’ by Anton Chekhov

‘My Mother Never Worked’ by Bonnie Smith Yachel

‘Keeping Errors At Bay’ by Bertrand Russell

Unit: 3 Conducting Meeting**[5 Hrs]**

- Introduction
- Writing notice
- Writing agenda
- Writing minutes

Unit: 4 Writing**[9hrs]**

- Fundamental of effective writing: Unity, Length, Coherence, Cohesion, Conciseness, and clarity.
Technical report writing: Format for technical report, Writing technical reports
- Proposal writing: Format for technical proposals, Writing technical proposals
- Writing letters:
Official letters: letters of request, complaint, apology and order
Job application with resume
- Writing memoranda

Unit: 5 Elements of Structures and Varieties in English Language**[11Hrs]**

Varieties in English languages, Preposition; subject- verb arrangement; Voice, reported speech

Reference Books:

1. Adhikari, U; Yadav, R; Shrestha, R.N. Communicative Skills in English, Research Training Unit, Department of Science and Humanities, Institute of Engineering, Pulchowk Campus 2002.
2. Anne Eisenberg, Effective Communication, Mc. Graw-Hill, 1982.
3. K.WHoup and T.E Pearsall, Reporting Technical Information, 5th Edition Macmillian Publishing Company, New York, 1984.
4. Leech, G, Savartivk, J., A Communicative Grammar of English, ELBS 1975
5. Oxford Advanced Learner's Dictionary of Current English, New Edition, Oxford.

Sociology and Professional Ethics
BT106HS

Year: I

Semester: I

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1		Theory	Practical	Theory	Practical	100
			40		60	-	

Course Objective:

The course aims at providing the basic knowledge on the concept of sociology and professional ethics. Its main purpose is to make students understand the link among sociological, cultural, economic, political aspects and technology and enable students apply technology to make better society.

Course Contents:

Unit 1: Introduction

[4Hrs]

- a. Definition and evolution of Sociology
- b. Relationship of sociology with Economic, Political science and Computer science
- c. Application of sociology in identification, implementation and evaluation of social and technical issues.

Unit 2: Social and Cultural change

[7Hrs]

- a. Process
- b. Theories of social change (evolutional, functional, conflict)
- c. Factors of Social change (economic, technology, education and demography)
- d. Role of media and communication in social and cultural change
- e. Resistance of social change
- f. Technological changes and its consequences

Unit 3: Characteristics of Nepali Society and Culture

[8 Hrs]

- a. Historical development of Nepal
- b. Demographic Composition
- c. Issue of Gender
- d. Caste and Ethnic group
- e. National integration and differentiation

- f. Social Stratification
- g. Social Problems & Controls

Unit 4: Ethical Issues in IT

[7Hrs]

- a. Definition of Profession
- b. Professional Ethics
- c. Code of conduct
- d. Ethical dilemma and problems
- e. Disciplinary action
- f. Corporation social responsibility

Unit 5: Process of Transformation

[6Hrs]

- a. Modernization
- b. Globalization
- c. Migration
- d. E-governance
- e. E-commerce

Unit 6: Understanding Development

[6Hrs]

- a. Definition and approaches of development
- b. Indicators of development
- c. Features of developing countries
- d. Development Planning
- e. Role of national and international community and state

Unit 7: Introduction to Emotional Intelligence

[7Hrs]

- a. Definition and benefits of Emotional Intelligence
- b. Components of Emotional Intelligence
- c. Role of Emotional Intelligence at social environment
- d. Emotional leadership development
- e. Skills in Emotional Intelligence: Emotions, Use emotions to facilitate thinking
- f. Optimism, Pessimism and the balance between Optimism & Pessimism

Reference Books:

1. Alex Inkles: "What is Sociology? Introduction in the discipline and profession
2. G.M. Foster: "Traditional Culture and Impacts of Technological Change"
3. Rishikeshav Raj Regmi, "Dimension of Nepali Society and Culture"
4. C.N.S.Rao, "Principle of Sociology with an introduction of Social Thought", S.Chand
5. Pratley Peter, "The Essence of Business Ethics"
6. A.Giddens&D,Mitchell, "Introduction to Sociology"

Artificial Intelligence & Intelligent Systems
BT152CO

Year: I

Semester: II

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1	2	Theory	Practical	Theory	Practical	100
			20	20	60	-	

Course Objective:

The main objective of this course is to familiarize students with the understanding of advanced AI technologies and a practical foundation in AI and its applications.

Course Contents:

Unit 1: Introduction

[2 Hrs]

- 1.1 Definition of Intelligent System
- 1.2 Types of Artificial Intelligent Systems
- 1.3 Overview of classical and contemporary Intelligent Systems

Unit 2: Inference and Reasoning

[14 Hrs]

- 2.1 Inference Rules
- 2.2 Horn clauses
- 2.3 Resolution in FOPL
- 2.4 Resolution Refutation System (RRS), Answer extraction from RRS, Rule based deduction system
- 2.5 Forward and backward reasoning
- 2.6 Reasoning and Uncertainty, Monotonicity
- 2.7 Statistical Reasoning, Bayesian Network
- 2.8 Case-based reasoning
- 2.9 Case Study

Unit 3: Expert System

[11 Hrs]

- 3.1 Introduction to Learning in Machines
- 3.2 Architecture and characteristics of an expert system

- 3.3 Categories of knowledge, Knowledge acquisition, Knowledge elicitation techniques
- 3.4 Forward chaining vs Backward chaining
- 3.5 Development of an expert system
- 3.6 Case Study

Unit 4: Neural Networks

[11 Hrs]

- 4.1 Basic concepts of Neuron, Network structure
- 4.2 Perceptron, Multilayer Perceptron, Adaline, Madaline
- 4.3 Back propagation
- 4.4 Learning in Simple Neuron, Deep learning
- 4.5 Hopfield network, Boltzmann Machines
- 4.6 Applications of Neural Network
- 4.7 Case Study

Unit 5: Natural Language Processing

[7 Hrs]

- 5.1 Introduction and Components of NLP
- 5.2 Steps in NLP: Morphological analysis, Syntactic analysis, Semantic analysis, Discourse integration, Pragmatic analysis
- 5.3 Parsing and Parse-tree generation
- 5.4 Case Study

Laboratory Work:

There shall be lab exercises on PROLOG/LISP covering the following topics.

1. Solving complex relation problems
2. Solving complex numerical problems such as Prime number, Fibonacci Series
3. Complex Logic-circuit programs
4. Menu-driven programs
5. Inference and reasoning problems
6. Tower of Hanoi

Reference Books:

1. Stuart Russel & Peter Norvig, "*Artificial Intelligence: A Modern Approach*", Pearson
2. E. Rich & K. Knight, "*Artificial Intelligence*", McGraw-Hill
3. D. W. Patterson, "*Artificial Intelligence and Expert Systems*", Prentice Hall
4. R. Shingal, "*Formal Concepts in Artificial Intelligence*", Chapman & Hall
5. E. Turban, "*Decision Support and Expert System*", Macmillan
6. G. Gazadar & C. Mellish, "*Natural Language Processing in Prolog and Introduction to Computational Linguistics*", Addison-Wesley
7. Haykin, "*Neural Networks: A Comprehensive Fundamentals*", Macmillan
8. D. Crookes, "*Introduction to Programming in Prolog*", Prentice Hall

Microprocessors and Assembly Language
BT153CO

Year: I

Semester: II

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1	2	Theory	Practical	Theory	Practical	100
			20	20	60	-	

Course Objective:

To become familiar with the operation, programming and application of microprocessors.

Course Contents:

Unit 1: Introduction

[3 hrs]

Microcomputer, Block diagram of Microcomputer, Microprocessor, Overview of Microprocessor family.

Unit 2: Intel 8085 Microprocessor

[14 hrs]

Pin Diagram and Pin Functions, Internal Architecture, Addressing Mode, Instruction Set with classification, Instruction format and programming, Timing Diagram.

Unit 3: I/O Interface

[6 hrs]

Introduction, I/O port addressing, Decoding, Serial and Parallel communication, The 8255 Programmable Peripheral Interface (Block diagram and Modes), The 8251 (Block diagram and Modes).

Unit 4: Interrupts

[2 hrs]

Introduction, Types of Interrupts in 8085/8086, Basic Interrupt Processing.

Unit 5: Memory Interface

[2 hrs]

Introduction, Address Decoding, 8088 Memory Interface.

Unit 6: DMA

[1 hr]

Introduction, Basic DMA Operation.

Unit 7: 8086 Instruction Description and Assemble Directives

[14 hrs]

Pin Diagram and Pin Functions, Internal Architecture, Addressing Mode, Instruction Set with classification and Programming.

Unit 8: Introduction to Higher Series of Intel Processors, A Comparative Study. [3 hrs]

Reference Books:

1. Barry B. Brey, The INTEL Microprocessors 8086/8088, 80186, 80286, 80386 and 80486 (Architecture, Programming and Interfacing), PHI
2. Yu Chung Liu & G. A. Gibson, Microcomputer Systems: The 8086/8088 Family Architecture, Programming and Design, EE Edition
3. Adam Osborne & J. Kane, An Introduction to Microcomputer, Vol. II - Some Real Microprocessors, Galgotia Book Source, New Delhi
4. Douglas V. Hall, Microprocessor & Interfacing, Programming and Hardware, Tata Mc-Graw Hill

Laboratory Work:

All Laboratory work will be based on assembler (8086). Before starting lab exercises, students should verify all the basic instructions.

Laboratory 1

To enter and to trace a simple assembly language program using DOS DEBUG routine.

Laboratory 2

To write a simple assembly language program that will make logical decisions based on program data, DOS DEBUG routine.

Laboratory 3

To enter and trace a program that contains a loop e.g. the loop instruction to use CX register as a default index counter and decrement CX at the end of each pass, compare CX to zero and if it is greater than zero, jump to the beginning of the loop.

Laboratory 4

To modify the program from laboratory 3, so that the looping is accomplished in a different way. Modify the program so that it does not rely on the 'loop' instruction, but rather, performs the loop operations separately. The output of laboratory 3 and laboratory 4 should be same.

Laboratory 5

To complete an assembly language program to carry out given logical process and then assemble, link and run. For example, to complete a given program so that it will read a single character from the keyboard and test the character and based on the range within which it falls, either print it, or change it to lower case and print it, or print it unmodified.

Laboratory 6

To modify the program of laboratory 5, so that it will read a string of characters and select them by the same as in laboratory 5.

Probability and Statistics
BT154CO

Year: I

Semester: II

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1		Theory	Practical	Theory	Practical	100
			40		60	-	

Course objective:

This course will provide a basic high- level introduction to the probability and statistics that underpin many of the modern machine learning and AI algorithms.

Course contents:

Unit 1:Introduction and descriptive statistics [7Hrs]

An overview of probability and statistics, describing data with graphs, measure of central tendency(mean, median, mode, partition values), measure of dispersion(Range, quartile deviation, standard deviation, variance, coefficient of variation).

Unit 2:Correlation and Regression [6 hrs]

Basic concepts of correlation and regression, types and measurement of correlation by karlpearson's correlation coefficient method, properties , least square methods, measurement of slope and intercept, properties of regression coefficients.

Unit 3:Probability [6Hrs]

Random experiment, sample space trial and events, types of events, Permutation, combination, mathematical definition of probability, addition and multiplication law of probability, bayes' theorem. Random variable and mathematical expectation

Unit 4:Probability distribution [7 Hrs]

Introduction, binomial distribution, poisson distribution and normal distribution

Unit 5:Estimation [4 Hrs]

Concept of sample, population, census, statistic, parameter, type of estimation, confidence for a single mean, confidence for difference of two means, confidence for a single proportion, confidence for the difference of two proportions.

Unit 6: Testing of hypothesis**[9Hrs]**

Hypothesis test concerning: single mean, difference of two means, single proportion, difference of two means, single proportion, difference of two proportion, chi-square test for goodness of fit and independence of attributes.

Unit 7: Application of computer on statistical data computing**[6 Hrs]**

Analysis of data using Excel, SPSS

Reference Books:

1. Richard A. Johnson, "Probability and Statistics for engineers 7th edition, Miller and Freund's publication.
2. Richard Levin, David S Rubin, statistics for management prentice hall publication
3. C. Gupta, "Fundamental of Statistics", Himalaya Publishing House.

Project I
BT156CO

Year: I

Semester: II

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
		3	Theory	Practical	Theory	Practical	100
				40		60	

Course Objective:

The aim of this course is to provide sound knowledge of linear algebra.

Course Contents:

Unit 1: Linear equations in linear algebra

[8 Hrs]

- 1.1 System of linear equations.
- 1.2 The matrix equation $Ax = b$.
- 1.3 Solution sets of linear system, Gaussian elimination and Gauss-Jordan reduction, rank Of matrix.
- 1.4 Linear independence .

Unit 2: Matrix Algebra

[7Hrs]

- 2.1Matrix operation.
- 2.2 The inverse of matrix.
- 2.3Solution of linear systems.
- 2.4 Characterizations of invertible Matrices.
- 2.5Application to computer graphics.

Unit 3: Determinants

[5 Hrs]

- 2.3 Introduction to determinants.
- 2.4 Properties of determinants.
- 2.5 Cramer's rule , volume and linear transformations.

Unit 4: Vector spaces

[11Hrs]

- 4.1 Vector spaces and sub spaces.
- 4.2 null spaces ,column spaces and linear transformations.
- 4.3 Linearly independent sets ,Bases.
- 4.4 Co-ordinate systems

4.5 The dimension of a vector space ,Rank

4.5 Change of basis.

Unit 5: Eigen values and Eigen vectors

[7Hrs]

5.1 Eigen values and Eigen vectors

5.2 The characteristic equation

5.3 Diagonalization

5.4 Eigenvectors and linear transformation.

Unit 6: Orthogonality and least squares

[7Hrs]

6.1 Inner product, length and orthogonality.

6.2 Orthogonal sets , orthogonal projections.

6.3 The Gram-schmidt process, least square problems

Reference Books:

1.Linear algebra and its applications , Third edition, David C. Lay
(Dorling kinders ley india Pvt. Ltd

Project 1
BT156CO

Year: I

Semester: II

Teaching Schedule Hours/Week			Examination Scheme				
Theory	Tutorial	Practical	Internal		Final		Total
3	1		Theory	Practical	Theory	Practical	100
			40		60	-	

To design and complete the software project using Python or design and implement an Intelligent System. On the completion of the project, student will be able to develop small scale software in Python programming language or build an intelligent System.

Course Contents:

There should be a total of 45 hours covering important features of object-oriented programming. A software development project will be assigned to students in a group (upto 3). A relevant topic shall be identified and instructed to each group. Students must develop the assigned software, submit written report, and give oral presentation.

General Procedure:

1. Topic Selection
2. Information Gathering
3. System Requirements and Specifications
4. Algorithms and Flowcharts
5. Coding
6. Implementation
7. Documentation

The project document shall include the following:

1. Technical description of the project
2. System aspect of the project
3. Project tasks and time-schedule
4. Project team members
5. Project supervisor
6. Implementation of the project

Project Evaluation Criteria for Internal assessment:

The marks allocated for the project should be evaluated based on the following criteria:

- Title identification and Proposal Writing— 10 Marks
- Mid-term Presentation — 20 Marks
- Pre-final Submission and final Presentation — 30 Marks

Project Evaluation Criteria for External assessment:

The marks allocated for the project should be evaluated based on the following criteria:

- Project Documentation— 20 Marks
- Final Presentation — 10 Marks
- VIVA--- 10 Marks

